

Economics G6410 - Mathematical Methods for Economists - Fall 2021
Math Camp Exam

Department of Economics
Columbia University

Instructions:

This is a timed closed book exam. You have 3 hours to solve the exam and return a scanned, typed or a hard copy of your answers to the TA (and cc the instructor). In case, you choose to email your answers, please ensure that you send us a single PDF document. There are 5 questions in total and the full grade is 100 points. You are not allowed to discuss your answers with any other students. In case of any clarifications, explicitly state the assumptions you are making to solve the question. Kindly show your complete work. Partial credit would be awarded in case of incomplete answers. No credit would be awarded if working is not shown. Columbia's academic honor code applies.

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1. **(40 points (5 points each))** Are the following statements true or false? If true, provide a proof. If false, provide a counter example
 - (a) Let (X, d) be a non-empty metric space, and $S \subset X$. If S has a limit point, then S has an infinite number of elements.
 - (b) Every diagonalizable matrix is invertible.
 - (c) Every convergent sequence is a Cauchy sequence.
 - (d) Let (X, d_X) and (Y, d_Y) be metric spaces, and function $f : X \rightarrow Y$. The function f is a continuous function iff $f(E)$ is open in (Y, d_Y) for any set E open in (X, d_X) .
 - (e) Let (X, d) be a metric space and $S \subset X$. If S is closed and bounded then it is compact.
 - (f) Let (X, d) be a complete metric space, T_1 and T_2 be self-maps on X . If T_1 is a contraction, and T_2 is non-expansive, i.e., $d(T_2(x), T_2(y)) \leq d(x, y)$ for any $x, y \in X$, then both composite self-maps $T_1 \circ T_2$ and $T_2 \circ T_1$ are contractions.
 - (g) An arbitrary intersection of compact sets is compact.
 - (h) Let $\{x_n\}_{n=1}^\infty$ and $\{y_n\}_{n=1}^\infty$ be two convergent sequences in $\mathbb{R}$ such that $x_n > y_n$ for all $n \in \mathbb{N}$ and $x_n \rightarrow x$ and $y_n \rightarrow y$. Then $x > y$.
2. **(15 points)** Consider the following matrix

$$A = \begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix}$$

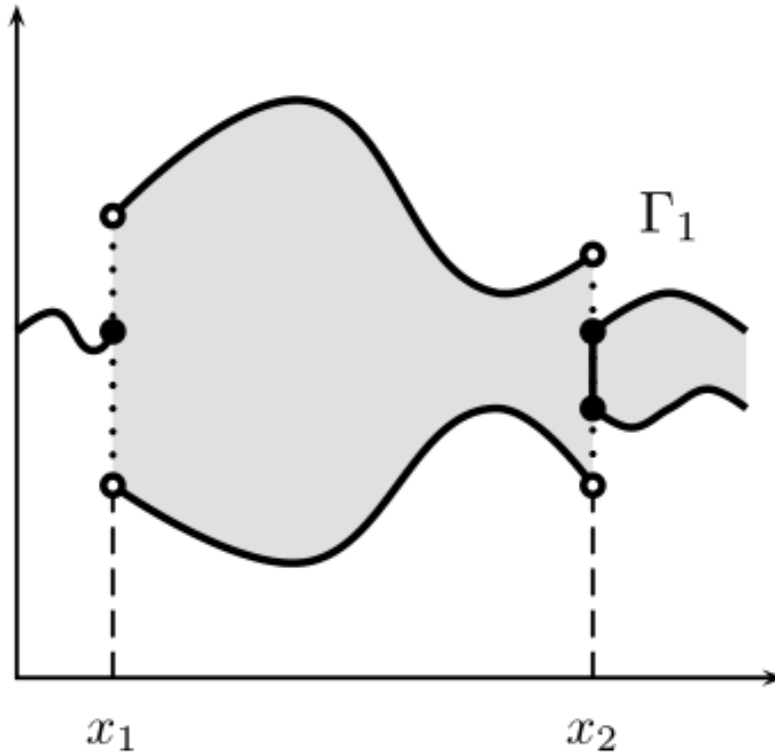
is it diagonalizable in $\mathbb{C}$? Is it diagonalizable in $\mathbb{R}$? If yes, diagonalize it, i.e. find a 2×2 matrix P such that $A = P\Lambda P^{-1}$, where Λ is a diagonal matrix. Also, find an expression for A^n for an $n \in \mathbb{N}$. As $n \rightarrow \infty$, does A^n converge to some real matrix B ? Is the matrix A positive definite? negative definite? neither?

3. **(15 points (5 points each))**. Consider the following function defined for $X_1 > 0, X_2 > 0, \dots, X_n > 0$ as

$$Y = F(X_1, X_2, \dots, X_n) = \sum_{i=1}^n \alpha_i X_i^\gamma$$

where $\alpha_i > 0, i = 1, \dots, n$ and $\gamma \neq 0$ are parameters.

- Is F homogeneous? If so, state its order of homogeneity. Is F homothetic?
 - For what values of γ is F concave? For what values of γ is it convex?
(Hint: Consider three different cases of γ , $\gamma < 0$, $\gamma \in (0, 1]$, $\gamma \geq 1$) and evaluate the second order derivative of x_i^γ for each case. What would this tell you? Is the sum of concave functions a concave function?
 - Log-linearize the equation $Y = F(X_1, X_2, \dots, X_n)$ around a point $(Y^*, X_1^*, \dots, X_n^*)$ satisfying this equation and represent the linearised equation in a percentage deviation format, i.e., the variables should take the form $\hat{z} = \frac{z - z^*}{z^*}$.
4. **(10 points)** For the correspondence F pictured below, answer the following questions. No formal proof required, a brief justification would suffice.



- (3 points)** Is the correspondence upper hemicontinuous at x_1 and x_2 ?
- (3 points)** Is the correspondence lower hemicontinuous at x_1 and x_2 ?
- (3 points)** Is the correspondence closed valued? compact valued? convex valued?
- (1 points)** Does the correspondence have a closed graph?

5. **(20 points)** Consider the following problem

$$\max_{(x_1, x_2) \in \mathbb{R}^2} x_1^{\frac{1}{2}} + x_2^{\frac{1}{2}}$$

s.t

$$\begin{aligned} p_1 x_1 + p_2 x_2 &\leq m, \\ x_1 &\geq 0, \\ x_2 &\geq 0 \end{aligned}$$

where $p_1, p_2, m \in \mathbb{R}_{++}$ are parameters.

- (a) **(4 points)** Does a solution exist? If yes, why?
- (b) **(2 points)** Argue that the budget constraint binds at any possible maximum.
- (c) **(10 points)** Set up the Lagrange and solve the problem.
- (d) **(2 points)** Are there any potential maximisers where the constraint qualification would fail?
- (e) **(2 points)** Differentiate the indirect utility function w.r.t. m and your Lagrange function w.r.t. m to verify the envelope theorem.